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base. We will create a database, add tables, insert data, and perform various queries.

1. Create a Database:

```
CREATE DATABASE company;
```

2. Use the Database:

```
USE company;
```

3. Create a Table for Employees:

```
CREATE TABLE employees (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    email VARCHAR(100),  
    hire_date DATE  
);
```

4. Insert Data into the Employees Table:

```
INSERT INTO employees (first_name, last_name,  
    email, hire_date)  
VALUES ('John', 'Doe', 'john.doe@example.  
com', '2023-06-01'),  
    ('Jane', 'Smith', 'jane.smith@example.  
com', '2022-03-15'),  
    ('Emily', 'Johnson', 'emily.johnson@  
example.com', '2021-11-30');
```

5. Query Data from the Employees Table:

```
SELECT * FROM employees;  
  
SELECT first_name, last_name FROM employees  
WHERE hire_date > '2023-01-01';
```

6. Update Data in the Employees Table:

```
UPDATE employees  
SET email = 'john.newemail@example.com'  
WHERE id = 1;
```

7. Delete Data from the Employees Table:

```
DELETE FROM employees WHERE id = 1;
```

8. Create a Table for Departments:

```
CREATE TABLE departments (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(50),  
    location VARCHAR(50)  
);
```

9. Insert Data into the Departments Table:

```
INSERT INTO departments (name, location)  
VALUES ('HR', 'New York'),  
    ('Engineering', 'San Francisco'),  
    ('Marketing', 'Chicago');
```

10. Join Tables to Retrieve Data:

```
SELECT employees.first_name, employees.last_  
name, departments.name AS department  
FROM employees  
JOIN departments ON employees.department_id  
= departments.id;
```

Industry Trends and Applications

SQL continues to be a critical skill in various industries due to its efficiency in managing and querying relational databases. Here are some current trends and applications of SQL:

Data Warehousing:

SQL is extensively used in data warehousing, where large volumes of data are stored and analyzed. Data warehouses enable businesses to perform complex queries and generate reports for decision-making. Tools like Amazon Redshift and Google BigQuery leverage SQL for querying large datasets efficiently.

Business Intelligence (BI):

SQL plays a vital role in business intelligence platforms, where it is used to extract, transform, and load (ETL) data from various sources. BI tools like Tableau, Power BI, and Looker use SQL to create dashboards and visualizations that provide insights into business performance.

Data Analytics:

SQL is the go-to language for data analysts who need to retrieve and analyze data from relational databases. SQL's ability to handle complex queries and aggregate functions makes it ideal for data analysis tasks. Analysts use SQL to generate reports, identify trends, and support data-driven decision-making.

Web Development:

SQL is a fundamental skill for web developers who need to interact with databases to store and retrieve user data. Content management systems (CMS) like WordPress and e-commerce platforms like Magento rely on SQL databases to manage content and transactions.

Cloud Databases:

With the rise of cloud computing, SQL has become essential for managing databases hosted on cloud platforms. Services like Amazon RDS, Azure SQL Database, and Google Cloud SQL provide scalable and managed database solutions that use SQL for data management.

Big Data Integration:

SQL is increasingly being integrated with big data technologies